

MorphGrower: A Synchronized Layer-by-layer Growing Approach for Plausible Neuronal Morphology Generation

Nianzu Yang, Kaipeng Zeng, Haotian Lu, Yexin Wu, Zexin Yuan, Danni Chen, Shengdian Jiang, Jiaxiang Wu, Yimin Wang, Junchi Yan

ThinkLab-SJTU

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MorphGrower: Title, Authors, and Context

- Topology-aware, synchronized layer-by-layer CVAE (conditional variational autoencoder).
- Joint sibling-branch generation for **tree consistency**
- Bifurcation-only topology; growth snapshots for interpretability

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Motivation: Limits of Rule-based Methods and MorphVAE

- Rule-based approaches are slow, sensitive, and **cell-type specific**
- MorphVAE (variational autoencoder) weakly models inter-branch dependencies and yields **invalid multifurcations**
- Generation with real-tree references encourages **biologically grounded** growth

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Related Work Landscape: Rule-based vs MorphVAE vs MorphGrower

- Rule-based methods encode handcrafted priors and enforce **topological validity**
- MorphVAE uses sequence modeling on 3D walks with **long decoding issues**
- MorphGrower is progressive with sibling-pair CVAE for **stronger realism**

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Our Idea and Contributions: Layer-wise Sibling-Pair Generation

- Joint sibling generation captures dependencies and preserves **bifurcations only**
- Layer conditioning on prior growth encodes **developmental priors**
- Demonstrated **higher plausibility** and functional alignment than prior learning methods

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Tree Morphology Basics and Branch Layers

- Soma is the root, bifurcations have two children, and tips are leaves with **no offspring**
- Branches are paths between a multifurcation and a downstream tip or bifurcation
- BFS (breadth-first search)-ordered branch layers enable **synchronized generation** across the tree

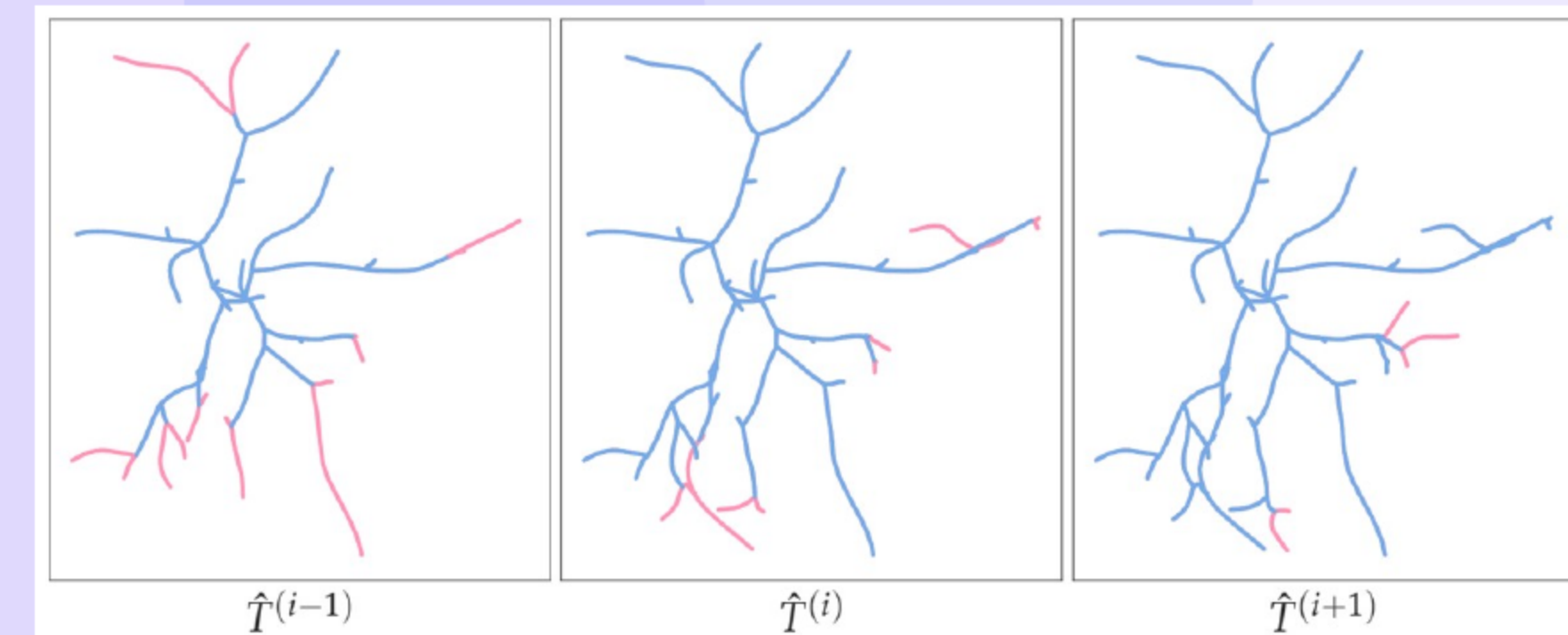
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Method Intuition: Why Sibling-Pair, Layer-wise Generation?

- Layer-wise strategy mirrors biological growth and reduces **one-shot complexity**
- Sibling pairs exhibit strong dependencies that benefit **joint generation**
- Prior growth informs local ancestors and the global forest for **coherent coverage**

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Growth Snapshots: Progressive Layer-wise Generation



- Consecutive plane snapshots highlight new layers in pink for **transparent growth**
- Intermediate trees reveal **stable expansion** without crossings
- Layer synchronization clarifies sibling-pair decisions at each step

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Qualitative Growth Micro-cases: Layer i to i+1 transitions

- Self-avoidance emerges from global conditioning with [smooth curvature](#)
 - Siblings bend into nearby gaps; avoid crossings near a crowded bifurcation
 - Paths stay smooth without sharp kinks
- Ancestor-informed orientation yields **balanced angles** and gentle turns
 - Initial tangents follow the parent curve; no sudden departures
 - Both children extend with gradual, symmetric turns
- Coverage balancing allocates siblings toward sparse sectors with [even spread](#)
 - One child fills the under-covered sector; the other stays near dense areas
 - Reduces visible holes without overlap

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Conditioning: Global Forest and Local Ancestors; von Mises–Fisher Latent

- Global context summarizes prior-layer forest with graph aggregation (GNN: graph neural network)
- Local context aggregates ancestor paths with temporal smoothing (LSTM: long short-term memory)
- Directional latent (vMF: von Mises–Fisher) and autoregressive decoder emit sibling branches with [stable shapes](#)

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Generation Protocol: Reference Layers and Soma Handling

- New morphologies grow auto-regressively by reference layer indices with **intermediate trees**
- Soma layer can be provided to stabilize [early orientation](#)
- Unconditional soma option exists but may **drift** global alignment

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Experimental Setup: Datasets and Baseline

- Four mouse datasets: VPM (ventral posteromedial thalamus), RGC (retinal ganglion cells), M1-EXC (motor-cortex excitatory), M1-INH (motor-cortex inhibitory); **standard splits**
- Baseline is MorphVAE (variational autoencoder) with tuned hyperparameters for **fair comparison**
- Embeddings are matched across methods to ensure [comparability](#)

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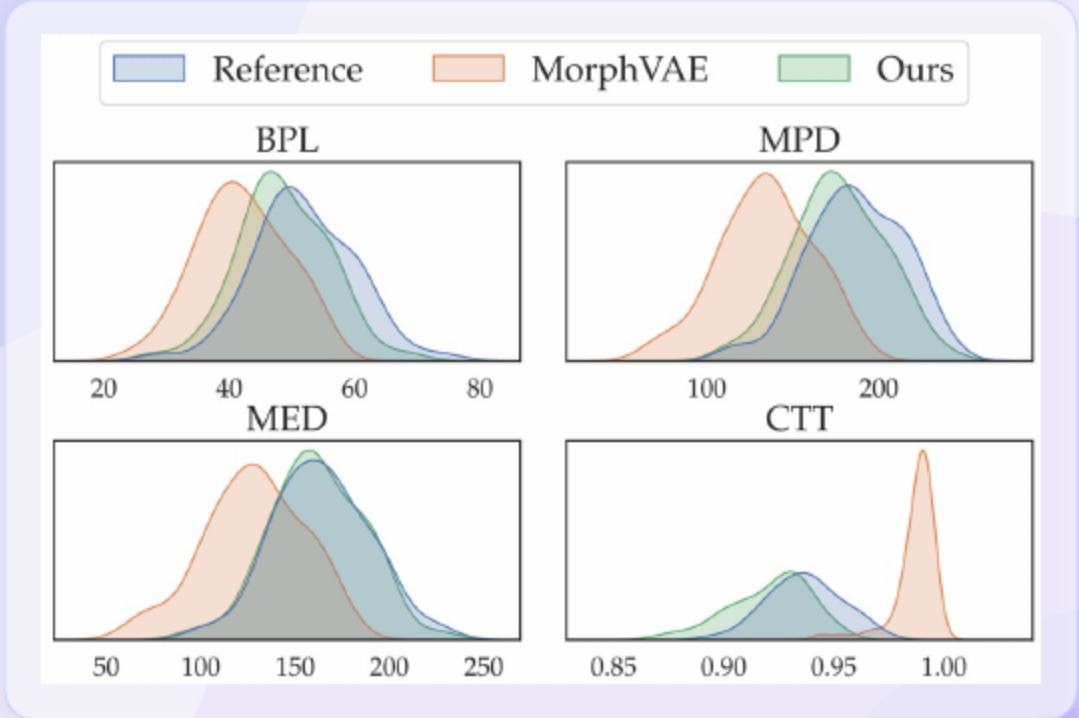
Datasets at a Glance (concise)

- VPM and RGC contain hundreds of reconstructions for **robust evaluation**
- M1-EXC includes hundreds while M1-INH includes tens with **limited scale**
- Cell-type context and sizes guide interpretation of **visuals and metrics**

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Metric Distributions: VPM (Qualitative Reading)

- VPM distributions (MPD, BPL, MED, CTT): MorphGrower closely tracks Reference with tighter histograms/CDFs.
- MorphVAE skews longer and appears less compact across these metrics.
- Branch-pair generation and branch-level resampling improve overall shape fidelity.



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Qualitative Case Studies: RGC and M1-EXC Visual Comparisons

- RGC examples show even sector coverage and **self-avoidance** with smooth flow
- M1-EXC results retain **balanced angles** and continuous trajectories
- MorphVAE tends toward **kinks** and branch crowding in deeper layers

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Real-vs-Fake Classifier Results

Dataset Method	VPM	RGC	M1-EXC	M1-INH
MorphVAE	86.75 ± 06.87	94.60 ± 01.02	80.72 ± 10.58	91.76 ± 12.14
MorphGrower	54.73 ± 03.36	62.70 ± 05.24	55.00 ± 02.54	54.74 ± 01.63

- Classifier accuracy remains near chance for MorphGrower indicating **high realism**
- MorphVAE classification remains **well above chance** across datasets
- Multi-view ResNet-18 CNN (convolutional neural network) blocks feed a linear head for **consistent assessment**

16 Whole-Arbor Qualitative: Coverage, Curvature, Crossings

- Even coverage: siblings fill sparse sectors without crowding
- Smooth curvature: no kinks; gradual turns across layers
- Fewer crossings/overlaps: global context steers branches into free space

17 Qualitative Recap: What Makes Outputs Look Real?

- Self-avoidance: siblings steer into free space to avoid crossings/overlaps at whole-arbor scale
- Smooth curvature: branches extend with gentle, continuous bends (no kinks)
- Balanced coverage: angles/lengths fill sparse regions without crowding dense sectors

18 Quantitative Snapshot: Metric Alignment (brief)

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- Across six L-measure metrics MorphGrower outperforms MorphVAE on **most comparisons**
 - Branch length and path distance show **clear gains** with pairing
 - Amplitude and orientation metrics remain **close to reference**

19 Functional Validation: Electrophysiology Alignment

- Averaged membrane potentials nearly overlap between real and generated with **low errors**
- Key waveform characteristics align across stimulation phases with **stable peaks**
- Functional similarity corroborates morphological **plausibility**

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Most Surprising Finding: Strong realism and functional match with minimal attributes

- Near-chance discrimination and matched electrophysiology occur **without diameters**
- Compact conditioning suffices for **visual realism** and function
- Minimal attributes still yield **robust generalization** across datasets

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Qualitative Panels: M1-INH Axonal Patterns and Coverage

- Collateral spacing appears plausible with **broad coverage** and smooth bends
- Loops and self-intersections are **avoided** under global context
- MorphVAE shows **zig-zag trajectories** and early terminations

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Failure vs Fix Micro-cases: Conditioning Resolves Conflicts

- Global and local conditioning prevent **crossings** by steering into free space
- Coverage gaps diminish through **balanced sector allocation**
- Ancestor alignment mitigates **orientation kinks** at departures

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Failure Modes and Limits: When MorphGrower Struggles

- Overly symmetric siblings and local crowding reduce **natural variability**
- Unconditional soma can **drift** orientation and deep layers can jitter
- Rare soma-edge motifs remain **underrepresented** without targeted data

- Sibling-pair generation, global context, and directional latent each provide **clear gains**
- Provided soma stabilizes **early layers** and global orientation
- Shorter resampling degrades smoothness with **rough trajectories**

- MorphGrower delivers **plausible topology-aware** growth and surpasses MorphVAE
- Future work will add diameters and model **asynchronous dynamics**
- Scalable generation can accelerate studies of **neural computation**

Thank You

Questions & Discussion